

Unit C Review
Multiple Choice Bank — 75 Questions

Question Set

MC 1. Which feature of DNA's double helix structure enables accurate information storage and replication?

- A. Parallel arrangement of sugar-phosphate backbones
- B. Complementary base pairing between strands
- C. Random base sequences along each strand
- D. Covalent bonds between base pairs

Correct answer: B

Rationale: Complementary base pairing (A-T and G-C) ensures accurate replication as each strand serves as a template for synthesizing its complement. The antiparallel (not parallel) arrangement positions bases for hydrogen bonding, not covalent bonding.

MC 2. What is the primary function of histone proteins in eukaryotic cells?

- A. To synthesize new DNA strands during replication
- B. To transport DNA between cellular compartments
- C. To package DNA into nucleosomes for compaction
- D. To break hydrogen bonds during transcription

Correct answer: C

Rationale: DNA wraps around histone proteins to form nucleosomes, which coil into higher-order structures for chromosome condensation. This allows 2 meters of human DNA to fit into a 10-micrometer nucleus while maintaining accessibility.

MC 3. A human somatic cell contains how many chromosomes?

- A. 23 chromosomes
- B. 46 chromosomes
- C. 92 chromosomes
- D. 22 chromosomes

Correct answer: B

Rationale: Human somatic cells are diploid (2n) and contain 46 chromosomes arranged as 23 pairs - 22 pairs of autosomes and 1 pair of sex chromosomes. Gametes contain 23 chromosomes (haploid).

MC 4. Why do sex cells maintain a haploid chromosome number?

- A. To double the genetic material with each generation
- B. To prevent chromosome numbers from doubling at fertilization
- C. To increase genetic redundancy in offspring
- D. To eliminate damaged chromosomes before reproduction

Correct answer: B

Rationale: Sex cells are haploid (n) to ensure that fertilization restores the diploid number rather than doubling chromosomes with each generation. This maintains constant chromosome numbers across generations.

MC 5. Which gene on the Y chromosome triggers male development in humans?

- A. XRY gene
- B. SRY gene
- C. TDF gene
- D. MDY gene

Correct answer: B

Rationale: The SRY gene on the Y chromosome triggers male development. Its absence results in female development through default developmental pathways, demonstrating how chromosome content determines biological outcomes.

MC 6. What happens to triploid plants during meiosis?

- A. Normal pairing of two homologous chromosomes
- B. Enhanced crossing over between chromosomes
- C. Inability to pair three homologous chromosomes properly
- D. Automatic doubling of chromosome sets

Correct answer: C

Rationale: Triploid plants cannot undergo normal meiosis because three homologous chromosomes cannot pair properly, resulting in sterile but viable organisms like seedless watermelons.

MC 7. What type of DNA structure is found in mitochondria and chloroplasts?

- A. Linear DNA with histone proteins
- B. Circular DNA molecules
- C. Fragmented DNA segments
- D. Double-stranded RNA molecules

Correct answer: B

Rationale: Mitochondria and chloroplasts maintain circular DNA molecules with bacterial-like structure, providing evidence for endosymbiotic origins and encoding essential components of electron transport chains.

MC 8. During which phase of the cell cycle does DNA replication occur?

- A. G1 phase
- B. S phase
- C. G2 phase
- D. M phase

Correct answer: B

Rationale: DNA replication occurs during S phase (synthesis), producing identical sister chromatids joined at centromeres. This follows G1 growth phase and precedes G2 preparation for mitosis.

MC 9. What is the result of semiconservative DNA replication?

- A. Both new DNA molecules contain only new strands
- B. One molecule has old strands, one has new strands
- C. Each new DNA molecule contains one old and one new strand
- D. Random distribution of old and new DNA segments

Correct answer: C

Rationale: Semiconservative replication produces two DNA molecules, each containing one original parental strand and one newly synthesized strand, ensuring accurate transmission of genetic information.

MC 10. Which checkpoint in the cell cycle verifies proper chromosome attachment to spindle fibers?

- A. G1/S checkpoint
- B. Spindle checkpoint
- C. G2/M checkpoint
- D. DNA damage checkpoint

Correct answer: B

Rationale: The spindle checkpoint during metaphase ensures all chromosomes attach to spindle fibers before proceeding to anaphase, preventing chromosome missegregation and aneuploidy.

MC 11. What distinguishes cytokinesis in plant cells from animal cells?

- A. Plant cells form a cleavage furrow
- B. Animal cells build a cell plate
- C. Plant cells construct a cell plate from vesicles
- D. Both use identical mechanisms

Correct answer: C

Rationale: Plant cytokinesis involves cell plate formation from Golgi-derived vesicles that fuse to create new cell walls, while animal cells form contractile rings creating cleavage furrows.

MC 12. During meiosis I, what type of chromosomes pair together?

- A. Sister chromatids
- B. Homologous chromosomes
- C. Non-homologous chromosomes
- D. Sex chromosomes only

Correct answer: B

Rationale: Homologous chromosomes (maternal and paternal versions) pair during meiosis I prophase, forming bivalents that enable crossing over and genetic recombination.

MC 13. What is the primary source of genetic variation during meiosis?

- A. DNA replication errors only
- B. Random fertilization only
- C. Crossing over and independent assortment
- D. Chromosome duplication

Correct answer: C

Rationale: Crossing over exchanges DNA segments between homologous chromosomes, while independent assortment randomly distributes maternal and paternal chromosomes, creating extensive genetic variation.

MC 14. How many possible chromosome combinations can human gametes have due to independent assortment alone?

- A. 23 combinations
- B. 46 combinations
- C. 529 combinations
- D. 8.4 million combinations

Correct answer: D

Rationale: With 23 chromosome pairs, independent assortment produces $2^{23} = 8.4$ million possible combinations in human gametes, not including additional variation from crossing over.

MC 15. What occurs during anaphase II of meiosis?

- A. Homologous chromosomes separate
- B. Sister chromatids separate
- C. Crossing over takes place
- D. Nuclear envelopes reform

Correct answer: B

Rationale: During anaphase II, sister chromatids separate and move to opposite poles, similar to mitotic anaphase. Homologous chromosomes separated earlier in anaphase I.

MC 16. In Mendel's monohybrid cross, what is the expected phenotypic ratio in the F₂ generation?

- A. 1:1
- B. 2:1

- C. 3:1
- D. 9:3:3:1

Correct answer: C

Rationale: Monohybrid crosses produce a 3:1 phenotypic ratio in F₂ generation (3 dominant : 1 recessive), while dihybrid crosses produce 9:3:3:1 ratios.

MC 17. What type of dominance produces a 1:2:1 phenotypic ratio in F₂ generation?

- A. Complete dominance
- B. Incomplete dominance
- C. Codominance
- D. Sex-linked inheritance

Correct answer: B

Rationale: Incomplete dominance produces intermediate phenotypes in heterozygotes, resulting in 1:2:1 ratio (1 homozygous dominant : 2 heterozygous : 1 homozygous recessive).

MC 18. Which blood type genotype demonstrates codominance?

- A. IAIA
- B. IBIB
- C. IAIB
- D. ii

Correct answer: C

Rationale: The IAIB genotype produces AB blood type, demonstrating codominance where both A and B antigens are equally expressed on red blood cells.

MC 19. What causes sex-linked traits to appear more frequently in males?

- A. Males have stronger gene expression
- B. Males have only one X chromosome
- C. Y chromosome carries more mutations
- D. Female hormones suppress trait expression

Correct answer: B

Rationale: Males are hemizygous for X-linked genes (one X chromosome), so recessive alleles are always expressed. Females need two recessive alleles for expression due to having two X chromosomes.

MC 20. According to Chargaff's rule, in DNA the amount of adenine equals the amount of:

- A. Guanine
- B. Cytosine
- C. Thymine
- D. Uracil

Correct answer: C

Rationale: Chargaff's rule states that adenine equals thymine and guanine equals cytosine due to complementary base pairing, providing crucial evidence for DNA's double helix structure.

MC 21. During transcription, what enzyme synthesizes RNA from a DNA template?

- A. DNA polymerase
- B. RNA polymerase
- C. Helicase
- D. Ligase

Correct answer: B

Rationale: RNA polymerase reads the DNA template strand (3' to 5') and synthesizes complementary RNA (5' to 3'), substituting uracil for thymine in the RNA sequence.

MC 22. What is the function of tRNA during translation?

- A. To unwind the DNA double helix
- B. To carry genetic information from nucleus to ribosome
- C. To deliver specific amino acids to the ribosome
- D. To catalyze peptide bond formation

Correct answer: C

Rationale: tRNA molecules carry specific amino acids to ribosomes, with anticodons complementary to mRNA codons ensuring proper amino acid sequence during protein synthesis.

MC 23. A frameshift mutation results from:

- A. Substitution of one nucleotide
- B. Insertion or deletion of nucleotides not in multiples of three
- C. Inversion of a chromosome segment
- D. Duplication of entire genes

Correct answer: B

Rationale: Frameshift mutations occur when insertions or deletions (not in multiples of three) shift the reading frame, altering all downstream codons and typically producing nonfunctional proteins.

MC 24. Which type of mutation changes a codon to a stop codon?

- A. Silent mutation
- B. Missense mutation
- C. Nonsense mutation
- D. Neutral mutation

Correct answer: C

Rationale: Nonsense mutations create premature stop codons, truncating proteins. Silent mutations don't change amino acids, while missense mutations substitute one amino acid for another.

MC 25. What is the function of restriction enzymes in genetic engineering?

- A. To replicate DNA molecules
- B. To cut DNA at specific recognition sequences
- C. To join DNA fragments together
- D. To transcribe DNA into RNA

Correct answer: B

Rationale: Restriction enzymes recognize and cut specific palindromic DNA sequences, creating sticky or blunt ends essential for recombinant DNA technology and genetic engineering applications.

MC 26. In PCR, what occurs during the annealing step?

- A. DNA strands separate at high temperature
- B. Primers bind to complementary sequences
- C. DNA polymerase extends new strands
- D. Products are analyzed by gel electrophoresis

Correct answer: B

Rationale: During annealing (50-65°C), primers bind to complementary sequences on single-stranded DNA templates. Denaturation separates strands, while extension synthesizes new DNA.

MC 27. What advantage do polyploid plants often exhibit?

- A. Smaller cell size

- B. Reduced stress tolerance
- C. Enhanced vigor and larger cell size
- D. Increased fertility

Correct answer: C

Rationale: Polyploid plants often show increased vigor, larger cell size, and enhanced stress tolerance, though odd-numbered polyploids typically experience meiotic difficulties and sterility.

MC 28. Why do triploid animals rarely survive to birth?

- A. They cannot undergo mitosis
- B. Precise gene dosage requirements cannot tolerate imbalanced expression
- C. They lack essential chromosomes
- D. Their cells are too large to function

Correct answer: B

Rationale: Triploid animals cannot survive because developmental pathways require precise gene dosage that cannot tolerate the imbalanced gene expression from having three copies of each chromosome.

MC 29. What type of chromosomal mutation involves transfer of a segment between non-homologous chromosomes?

- A. Deletion
- B. Duplication
- C. Inversion
- D. Translocation

Correct answer: D

Rationale: Translocation involves transfer of chromosome segments between non-homologous chromosomes. Deletion removes segments, duplication copies them, and inversion reverses segment orientation.

MC 30. In gel electrophoresis, DNA fragments separate based on:

- A. Their sequence composition
- B. Their size and charge
- C. Their temperature stability
- D. Their protein content

Correct answer: B

Rationale: Gel electrophoresis separates DNA fragments by size as they migrate through gel matrix under electric field, with smaller fragments moving faster than larger ones.

MC 31. What determines whether nondisjunction results in viable or lethal outcomes?

- A. The size of the cell
- B. The specific chromosome affected and gene content
- C. The age of the organism
- D. The presence of telomeres

Correct answer: B

Rationale: Viability depends on which chromosome is affected - sex chromosome aneuploidies are often viable while autosomal aneuploidies (except small chromosomes like 21) are typically lethal due to gene dosage imbalances.

MC 32. During prokaryotic gene regulation, what binds to the operator region?

- A. RNA polymerase
- B. Repressor protein
- C. Inducer molecule
- D. Ribosome

Correct answer: B

Rationale: Repressor proteins bind to operator regions to block RNA polymerase access and prevent transcription. Inducers can bind repressors to release them from operators, allowing transcription.

MC 33. What modification occurs to eukaryotic mRNA before translation?

- A. Addition of amino acids
- B. 5' cap and 3' poly-A tail addition
- C. Conversion to DNA
- D. Circular formation

Correct answer: B

Rationale: Eukaryotic mRNA undergoes post-transcriptional modifications including 5' methylguanosine cap and 3' poly-A tail addition, plus intron removal, before export from nucleus for translation.

MC 34. What is the primary function of telomeres?

- A. To initiate DNA replication
- B. To protect chromosome ends from degradation
- C. To separate sister chromatids
- D. To enhance gene expression

Correct answer: B

Rationale: Telomeres are repetitive sequences at chromosome ends that protect against degradation during replication. They shorten with each division, linking to cellular aging.

MC 35. In a test cross, an organism with unknown genotype is crossed with:

- A. A homozygous dominant individual
- B. A heterozygous individual
- C. A homozygous recessive individual
- D. Another unknown individual

Correct answer: C

Rationale: Test crosses use homozygous recessive individuals to determine unknown genotypes - if any recessive offspring appear, the unknown parent must be heterozygous.

MC 36. What causes linked genes to be inherited together more frequently than expected?

- A. They code for similar proteins
- B. They are located on the same chromosome
- C. They undergo simultaneous mutations
- D. They share the same promoter

Correct answer: B

Rationale: Linked genes are located on the same chromosome and tend to be inherited together unless separated by crossing over. Recombination frequency indicates their distance apart.

MC 37. Which process produces recombinant chromosomes during meiosis?

- A. Independent assortment only
- B. Random fertilization
- C. Crossing over between homologous chromosomes
- D. Sister chromatid separation

Correct answer: C

Rationale: Crossing over during prophase I exchanges DNA segments between non-sister chromatids of homologous chromosomes, creating recombinant chromosomes with new allele combinations.

MC 38. What happens when lactose is present in bacterial cells with the lac operon?

- A. Repressor binds more tightly to operator
- B. Lactose binds repressor, allowing transcription
- C. RNA polymerase is degraded
- D. The operon is permanently deleted

Correct answer: B

Rationale: Lactose acts as an inducer, binding to the repressor protein and causing conformational change that releases it from the operator, allowing RNA polymerase to transcribe lac genes.

MC 39. During DNA replication, which enzyme relieves tension from unwinding?

- A. Helicase
- B. Primase
- C. Topoisomerase
- D. Ligase

Correct answer: C

Rationale: Topoisomerase relieves tension created by helicase unwinding the double helix. Primase synthesizes RNA primers, while ligase joins Okazaki fragments on the lagging strand.

MC 40. What distinguishes the leading strand from the lagging strand during DNA replication?

- A. Leading strand is synthesized continuously in 5' to 3' direction
- B. Lagging strand is synthesized continuously
- C. Leading strand requires multiple primers
- D. Lagging strand doesn't require primers

Correct answer: A

Rationale: The leading strand is synthesized continuously 5' to 3' toward the replication fork, while the lagging strand is synthesized discontinuously as Okazaki fragments away from the fork.

MC 41. In pedigree analysis, what pattern indicates autosomal recessive inheritance?

- A. Trait appears in every generation
- B. Only males are affected
- C. Trait skips generations and affects both sexes equally
- D. Only females can pass the trait

Correct answer: C

Rationale: Autosomal recessive traits typically skip generations (carriers don't express), affect both sexes equally, and can appear from unaffected carrier parents.

MC 42. What type of RNA carries the genetic code from DNA to ribosomes?

- A. tRNA
- B. rRNA
- C. mRNA
- D. snRNA

Correct answer: C

Rationale: mRNA (messenger RNA) carries genetic information from DNA in the nucleus to ribosomes in the cytoplasm for translation into proteins.

MC 43. Which codon signals the start of translation?

- A. UAA
- B. AUG
- C. UGA
- D. UAG

Correct answer: B

Rationale: AUG is the universal start codon that codes for methionine and initiates translation. UAA, UGA, and UAG are stop codons that terminate translation.

MC 44. What produces 'sticky ends' useful for genetic engineering?

- A. Restriction enzymes making staggered cuts
- B. DNA polymerase errors
- C. Blunt cutting by all enzymes
- D. Random DNA breakage

Correct answer: A

Rationale: Many restriction enzymes make staggered cuts in palindromic sequences, creating single-stranded overhangs (sticky ends) that can base-pair with complementary sequences.

MC 45. In CRISPR-Cas9, what guides the Cas9 enzyme to specific DNA sequences?

- A. Protein recognition domains
- B. Guide RNA complementary to target
- C. Random binding events
- D. Restriction enzyme sites

Correct answer: B

Rationale: Guide RNA with complementary sequence to the target DNA directs Cas9 to specific genomic locations for precise cutting, enabling targeted genetic modifications.

MC 46. What cellular structure is visible only during cell division?

- A. Nuclear envelope
- B. Condensed chromosomes
- C. Ribosomes
- D. Endoplasmic reticulum

Correct answer: B

Rationale: Condensed chromosomes become visible during cell division as chromatin condenses. During interphase, chromatin is dispersed and chromosomes are not individually visible.

MC 47. During which meiotic stage does synapsis occur?

- A. Prophase II
- B. Metaphase I
- C. Prophase I
- D. Anaphase I

Correct answer: C

Rationale: Synapsis (pairing of homologous chromosomes) occurs during prophase I of meiosis, forming bivalents that enable crossing over and genetic recombination.

MC 48. What explains the 9:3:3:1 phenotypic ratio in dihybrid crosses?

- A. Gene linkage
- B. Independent assortment of two gene pairs
- C. Incomplete dominance
- D. Sex-linked inheritance

Correct answer: B

Rationale: The 9:3:3:1 ratio results from independent assortment of two gene pairs on different chromosomes, with each pair showing complete dominance.

MC 49. Which enzyme joins Okazaki fragments on the lagging strand?

- A. Helicase
- B. Primase
- C. DNA ligase
- D. Topoisomerase

Correct answer: C

Rationale: DNA ligase catalyzes phosphodiester bond formation between adjacent Okazaki fragments after primer removal, creating a continuous lagging strand.

MC 50. What determines the reading frame during translation?

- A. The stop codon position
- B. The start codon AUG position
- C. The poly-A tail
- D. The 5' cap

Correct answer: B

Rationale: The start codon AUG establishes the reading frame by determining where translation begins, setting the three-nucleotide codon boundaries for the entire sequence.

MC 51. In incomplete dominance, what phenotype do F1 heterozygotes display?

- A. The dominant phenotype only
- B. The recessive phenotype only
- C. An intermediate phenotype
- D. Both parental phenotypes simultaneously

Correct answer: C

Rationale: Incomplete dominance produces intermediate phenotypes in heterozygotes because neither allele is completely dominant, resulting in blended expression.

MC 52. What phase follows cytokinesis in actively dividing cells?

- A. S phase
- B. G2 phase
- C. G1 phase
- D. M phase

Correct answer: C

Rationale: G1 phase follows cytokinesis, where cells grow, accumulate nutrients, and synthesize enzymes needed for DNA replication before entering S phase.

MC 53. Which statement describes homologous chromosomes?

- A. They are identical copies joined at centromeres
- B. They carry the same genes but may have different alleles
- C. They are found only in gametes
- D. They separate during metaphase II

Correct answer: B

Rationale: Homologous chromosomes are matching pairs (maternal and paternal) that carry the same genes at corresponding loci but may have different alleles for those genes.

MC 54. What prevents most aneuploid conditions in autosomes from being viable?

- A. Lack of centromeres
- B. Gene dosage imbalances affecting essential functions
- C. Inability to undergo mitosis
- D. Absence of telomeres

Correct answer: B

Rationale: Autosomal aneuploidies create severe gene dosage imbalances that disrupt essential cellular functions, making most lethal except for small chromosomes like chromosome 21.

MC 55. During translation, peptide bonds form between:

- A. Adjacent nucleotides
- B. Amino acids at the ribosome
- C. tRNA molecules
- D. mRNA codons

Correct answer: B

Rationale: Peptide bonds form between amino acids at the ribosome's catalytic site, linking them into growing polypeptide chains during translation.

MC 56. What molecular tool allows creation of recombinant DNA?

- A. DNA polymerase alone
- B. Restriction enzymes and DNA ligase
- C. RNA polymerase only
- D. Ribosomes

Correct answer: B

Rationale: Restriction enzymes cut DNA at specific sites and DNA ligase joins fragments, enabling insertion of foreign DNA into vectors for recombinant DNA technology.

MC 57. How does chromatin remodeling affect gene expression?

- A. It permanently deletes genes
- B. It regulates DNA accessibility for transcription
- C. It changes the DNA sequence
- D. It prevents DNA replication

Correct answer: B

Rationale: Chromatin remodeling complexes and histone modifications regulate which genes remain accessible for transcription by controlling DNA packaging density.

MC 58. What characterizes a carrier for an autosomal recessive disorder?

- A. Homozygous for the recessive allele
- B. Heterozygous with one normal and one mutant allele
- C. Homozygous for the dominant allele
- D. Affected by the disorder

Correct answer: B

Rationale: Carriers are heterozygous, possessing one normal and one mutant allele. They don't express the disorder but can pass the recessive allele to offspring.

MC 59. What is the consequence of a nonsense mutation?

- A. No change in protein function
- B. Single amino acid substitution
- C. Premature protein termination
- D. Enhanced protein activity

Correct answer: C

Rationale: Nonsense mutations create stop codons, causing premature termination of translation and typically producing truncated, nonfunctional proteins.

MC 60. During PCR, what happens at 94-96°C?

- A. Primers anneal to template
- B. DNA strands denature and separate
- C. New DNA synthesis occurs
- D. Products are analyzed

Correct answer: B

Rationale: The high temperature (94-96°C) denatures DNA by breaking hydrogen bonds between base pairs, separating the double helix into single strands for replication.

MC 61. What enables genetic variation through sexual reproduction?

- A. Mitosis only
- B. Meiosis and random fertilization
- C. Binary fission
- D. Budding

Correct answer: B

Rationale: Meiosis generates genetic variation through crossing over and independent assortment, while random fertilization combines different gametes, multiplying variation possibilities.

MC 62. Which structure contains the actual catalytic activity in ribosomes?

- A. tRNA
- B. mRNA
- C. rRNA
- D. DNA

Correct answer: C

Rationale: rRNA (ribosomal RNA) within ribosomes has catalytic activity (ribozyme function) that catalyzes peptide bond formation during translation.

MC 63. What happens to sister chromatids during anaphase of mitosis?

- A. They pair with homologous chromosomes
- B. They remain joined and move together
- C. They separate and move to opposite poles
- D. They undergo crossing over

Correct answer: C

Rationale: During anaphase, sister chromatids separate at centromeres and move to opposite cell poles, ensuring each daughter cell receives identical genetic information.

MC 64. In bacteria, what structure contains the main genetic material?

- A. Linear chromosomes in nucleus
- B. Circular chromosome in nucleoid region
- C. Multiple linear chromosomes
- D. RNA only

Correct answer: B

Rationale: Bacteria have a circular chromosome located in the nucleoid region (not membrane-bound), plus possible plasmids, contrasting with eukaryotic linear chromosomes in nuclei.

MC 65. What determines whether a mutation is beneficial, neutral, or harmful?

- A. The size of the mutation
- B. The effect on protein function and fitness
- C. The chromosome where it occurs
- D. The age when it occurs

Correct answer: B

Rationale: Mutation effects depend on how they alter protein function and organism fitness in specific environments - the same mutation can be beneficial, neutral, or harmful in different contexts.

MC 66. During prophase I of meiosis, what forms between paired homologous chromosomes?

- A. Sister chromatids
- B. Synaptonemal complex
- C. Cell plate
- D. Nuclear envelope

Correct answer: B

Rationale: The synaptonemal complex forms between homologous chromosomes during synapsis in prophase I, facilitating crossing over and genetic recombination.

MC 67. What modification helps protect mRNA from degradation?

- A. Intron insertion
- B. 5' cap and 3' poly-A tail
- C. Codon optimization
- D. Ribosome binding

Correct answer: B

Rationale: The 5' methylguanosine cap and 3' poly-A tail protect mRNA from exonuclease degradation, facilitate translation initiation, and regulate mRNA stability.

MC 68. In gel electrophoresis, why do smaller DNA fragments migrate faster?

- A. They have more negative charge
- B. They navigate through gel pores more easily
- C. They are more stable
- D. They bind more ethidium bromide

Correct answer: B

Rationale: Smaller DNA fragments move faster through gel matrix pores under electric field, while larger fragments are impeded, allowing size-based separation.

MC 69. What type of inheritance pattern shows continuous variation in traits?

- A. Simple dominance
- B. Codominance
- C. Polygenic inheritance
- D. Sex-linked inheritance

Correct answer: C

Rationale: Polygenic inheritance involves multiple genes contributing to one trait, producing continuous variation (like height or skin color) rather than discrete categories.

MC 70. Which checkpoint prevents cells with damaged DNA from entering mitosis?

- A. G1/S checkpoint
- B. G2/M checkpoint
- C. Spindle checkpoint
- D. Cytokinesis checkpoint

Correct answer: B

Rationale: The G2/M checkpoint verifies DNA integrity and proper replication before allowing entry into mitosis, preventing propagation of mutations.

MC 71. What enables bacteria to take up foreign DNA from their environment?

- A. Conjugation only
- B. Transformation when competent
- C. Binary fission
- D. Mitosis

Correct answer: B

Rationale: Transformation allows competent bacteria to take up naked DNA from their environment, incorporating it into their genome for new traits like antibiotic resistance.

MC 72. In humans, what determines biological sex at the chromosomal level?

- A. The number of X chromosomes only
- B. The presence or absence of Y chromosome
- C. Autosomal genes
- D. Mitochondrial inheritance

Correct answer: B

Rationale: The presence of Y chromosome (specifically SRY gene) triggers male development, while its absence results in female development through default pathways.

MC 73. What allows PCR to amplify specific DNA sequences?

- A. Random primer binding
- B. Sequence-specific primers flanking target
- C. Complete genome replication
- D. RNA polymerase activity

Correct answer: B

Rationale: PCR uses sequence-specific primers that flank the target region, allowing selective amplification of desired DNA sequences through repeated thermal cycles.

MC 74. During fertilization, what restores the diploid chromosome number?

- A. Mitosis of gametes
- B. Fusion of haploid gamete nuclei
- C. Chromosome duplication
- D. Meiosis II

Correct answer: B

Rationale: Fertilization combines two haploid gamete nuclei ($n + n = 2n$), restoring the diploid chromosome number characteristic of the species.

MC 75. What evidence supports the endosymbiotic origin of mitochondria?

- A. Linear DNA structure
- B. Circular DNA and bacterial-like structure
- C. Absence of DNA
- D. Nuclear membrane presence

Correct answer: B

Rationale: Mitochondria have circular DNA and bacterial-like structure, supporting endosymbiotic theory that they originated from engulfed bacteria that became organelles.

Unit C Review
Constructed Response Bank

Question 1

A researcher observes that human somatic cells contain 46 chromosomes while gametes contain 23 chromosomes.

- a. Explain why sex cells must maintain a haploid chromosome number.
- b. Describe how meiosis reduces chromosome number from diploid to haploid.
- c. State one consequence if nondisjunction occurs during meiosis I.

Question 2

A student claims that 'DNA replication is conservative, meaning the original DNA molecule remains intact while a completely new copy is made.'

- a. State whether this claim is correct or incorrect and explain the actual mechanism.
- b. Explain how complementary base pairing ensures accurate DNA replication.
- c. Identify one enzyme essential for DNA replication and its specific function.

Question 3

Research shows that crossing over during meiosis creates genetic variation in offspring.

- a. Describe when and how crossing over occurs during meiosis.
- b. Explain how crossing over contributes to genetic diversity beyond independent assortment.
- c. Calculate the number of possible gamete combinations in humans from independent assortment alone.

Question 4

A genetics counselor explains that cystic fibrosis is an autosomal recessive disorder affecting 1 in 2500 births.

- a. Explain what it means for a disorder to be autosomal recessive.
- b. Calculate the probability that two carrier parents will have an affected child.
- c. State why autosomal recessive disorders often skip generations in pedigrees.

Question 5

Scientists discover that histone modifications can regulate gene expression without changing DNA sequences.

- a. Explain how DNA organization around histones affects gene accessibility.
- b. Describe how histone modifications regulate transcription without altering DNA sequence.
- c. State one advantage of epigenetic regulation over permanent DNA changes.

Question 6

A biotechnology company uses restriction enzymes and DNA ligase to create recombinant DNA for insulin production.

- a. Explain how restriction enzymes create compatible DNA fragments for joining.
- b. Describe the role of DNA ligase in creating recombinant DNA molecules.
- c. Identify one advantage of producing human insulin in bacteria versus extracting from animal sources.

Question 7

A patient with Turner syndrome has 45 chromosomes instead of the normal 46.

- Explain which type of chromosomal abnormality causes Turner syndrome.
- Explain why sex chromosome aneuploidies are more viable than autosomal aneuploidies.
- State one physical characteristic commonly associated with Turner syndrome.

Question 8

Researchers observe that polymerase chain reaction (PCR) can amplify a single DNA molecule into billions of copies.

- Describe the three temperature-dependent steps of one PCR cycle.
- Explain why PCR produces exponential amplification of the target sequence.
- State why Taq polymerase is used instead of human DNA polymerase.

Question 9

A study finds that frameshift mutations typically have more severe effects than point mutations.

- Explain what causes a frameshift mutation and how it affects the reading frame.
- Compare the protein-level effects of frameshift versus missense mutations.
- State why insertion or deletion of exactly three nucleotides is less damaging.

Question 10

In snapdragons, red flowers crossed with white flowers produce pink flowers in the F₁ generation.

- Identify the type of dominance exhibited and explain how it differs from complete dominance.
- Predict the phenotypic ratio if two pink snapdragons are crossed.
- State the genotypic ratio from this cross.

Question 11

A molecular biologist states that 'The central dogma describes the flow of genetic information as DNA → RNA → Protein.'

- Explain the process by which DNA information is converted to RNA.
- Describe how RNA information becomes protein structure.
- Identify one exception to the central dogma's unidirectional flow.

Question 12

Seedless watermelons are triploid ($3n=33$) and cannot produce viable seeds.

- Explain why triploid organisms have difficulty undergoing meiosis.
- Describe how seedless watermelons are produced despite being sterile.
- State one advantage of polyploidy in crop plants.

Question 13

Research indicates that linked genes on the same chromosome do not assort independently during meiosis.

- Explain why genes on the same chromosome tend to be inherited together.
- Describe how recombination frequency relates to the distance between linked genes.
- State the maximum recombination frequency possible between two linked genes.

Question 14

A researcher discovers that telomeres shorten with each cell division, linking to cellular aging.

- a. Explain the structure and function of telomeres at chromosome ends.
- b. Describe why telomeres shorten with each replication cycle.
- c. Identify which enzyme can extend telomeres and where it is typically active.

Question 15

The lac operon in *E. coli* demonstrates how bacteria regulate gene expression in response to lactose availability.

- a. Explain how the lac operon is regulated when lactose is absent.
- b. Describe what happens to lac operon expression when lactose is present.
- c. State the advantage of inducible gene expression for bacterial cells.

Question 16

CRISPR-Cas9 technology allows scientists to make precise edits to DNA sequences in living cells.

- a. Explain how guide RNA directs Cas9 to specific DNA targets.
- b. Describe two ways cells can repair DNA after Cas9 creates a double-strand break.
- c. State one medical application of CRISPR-Cas9 technology.

Question 17

A genetic counselor notes that males are more frequently affected by X-linked recessive disorders like hemophilia.

- a. Explain why X-linked recessive traits affect males more than females.
- b. Predict the offspring ratios when a carrier female crosses with a normal male.
- c. State why an affected male cannot pass the disorder to his sons.

Question 18

Scientists observe that mitochondrial DNA is inherited exclusively from the mother in most organisms.

- a. Explain why mitochondrial inheritance follows a maternal pattern.
- b. Describe how mitochondrial DNA structure differs from nuclear DNA.
- c. State one application of mitochondrial DNA analysis in human studies.

Question 19

A pharmaceutical company wants to produce human growth hormone in bacterial cells for medical treatment.

- a. Explain why human genes must be modified before expression in bacteria.
- b. Describe how plasmid vectors facilitate gene transfer into bacterial cells.
- c. Identify one advantage of using bacteria for protein production.

Question 20

Researchers find that DNA packaging must be precisely regulated for proper cellular function.

- a. Explain how DNA compaction allows 2 meters of DNA to fit in a 10-micrometer nucleus.
- b. Describe the relationship between chromatin structure and gene expression.

c. State when chromosomes are most condensed during the cell cycle.

Model Answers & Rubrics

Question 1

Part a: Explain why sex cells must maintain a haploid chromosome number.

Model Answer: Sex cells (gametes) must be haploid ($n=23$) to prevent chromosome numbers from doubling with each generation. When two haploid gametes fuse during fertilization, they restore the diploid number ($2n=46$) characteristic of the species. If gametes were diploid, offspring would have 92 chromosomes, doubling with each generation. (2 marks) (2 marks)

Part b: Describe how meiosis reduces chromosome number from diploid to haploid.

Model Answer: Meiosis reduces chromosome number through two divisions. In meiosis I, homologous chromosomes pair and then separate, reducing chromosome number by half (46 to 23). Each daughter cell receives one chromosome from each homologous pair. Meiosis II then separates sister chromatids without further reduction in chromosome number. (2 marks) (2 marks)

Part c: State one consequence if nondisjunction occurs during meiosis I.

Model Answer: Nondisjunction during meiosis I results in gametes with abnormal chromosome numbers - some with $n+1$ and others with $n-1$ chromosomes, leading to aneuploid offspring like Down syndrome (trisomy 21). (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for explaining prevention of doubling, 1 mark for fertilization restoring diploid number

Part b (2 marks): 1 mark for describing meiosis I reduction, 1 mark for explaining homolog separation

Part c (1 mark): 1 mark for identifying aneuploidy or specific example

Question 2

Part a: State whether this claim is correct or incorrect and explain the actual mechanism.

Model Answer: The claim is incorrect. DNA replication is semiconservative, not conservative. Each new DNA molecule contains one original (parental) strand and one newly synthesized strand. During replication, the double helix unwinds and each strand serves as a template for a complementary new strand. (2 marks) (2 marks)

Part b: Explain how complementary base pairing ensures accurate DNA replication.

Model Answer: Complementary base pairing (A-T and G-C) ensures accuracy because each base can only pair with its complement through specific hydrogen bonding patterns. As DNA polymerase adds nucleotides, it matches adenine with thymine and guanine with cytosine, using the template strand to guide synthesis of an exact complementary copy. (2 marks) (2 marks)

Part c: Identify one enzyme essential for DNA replication and its specific function.

Model Answer: Helicase unwinds the double helix by breaking hydrogen bonds between base pairs, creating the replication fork where DNA synthesis occurs. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for identifying incorrect, 1 mark for explaining semiconservative replication

Part b (2 marks): 1 mark for base pairing rules, 1 mark for template function

Part c (1 mark): 1 mark for correct enzyme and function

Question 3

Part a: Describe when and how crossing over occurs during meiosis.

Model Answer: Crossing over occurs during prophase I of meiosis when homologous chromosomes pair up (synapsis) forming bivalents. Non-sister chromatids from homologous chromosomes exchange DNA

segments at points called chiasmata, swapping genetic material between maternal and paternal chromosomes. (2 marks) (2 marks)

Part b: Explain how crossing over contributes to genetic diversity beyond independent assortment.

Model Answer: Crossing over creates recombinant chromosomes with new combinations of alleles that didn't exist in either parent. While independent assortment shuffles whole chromosomes, crossing over creates new allele combinations within chromosomes, exponentially increasing genetic variation possibilities in gametes. (2 marks) (2 marks)

Part c: Calculate the number of possible gamete combinations in humans from independent assortment alone.

Model Answer: $2^{23} = 8.4$ million possible combinations from independent assortment of 23 chromosome pairs. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for prophase I timing, 1 mark for describing exchange process

Part b (2 marks): 1 mark for recombinant chromosomes, 1 mark for comparing to independent assortment

Part c (1 mark): 1 mark for correct calculation

Question 4

Part a: Explain what it means for a disorder to be autosomal recessive.

Model Answer: Autosomal recessive means the gene is located on an autosome (not sex chromosome) and two copies of the recessive allele are required for expression. Individuals with one recessive allele are carriers who don't express the disorder but can pass it to offspring. (2 marks) (2 marks)

Part b: Calculate the probability that two carrier parents will have an affected child.

Model Answer: Two carrier parents ($Aa \times Aa$) produce offspring with genotypes: 1 AA : 2 Aa : 1 aa. The probability of an affected child (aa) is 1/4 or 25%. The Punnett square shows 25% AA (unaffected), 50% Aa (carriers), and 25% aa (affected). (2 marks) (2 marks)

Part c: State why autosomal recessive disorders often skip generations in pedigrees.

Model Answer: They skip generations because carriers (heterozygotes) don't express the disorder, appearing unaffected while still passing the recessive allele to offspring. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for autosomal location, 1 mark for two alleles required

Part b (2 marks): 1 mark for correct probability, 1 mark for showing work/reasoning

Part c (1 mark): 1 mark for carrier explanation

Question 5

Part a: Explain how DNA organization around histones affects gene accessibility.

Model Answer: DNA wraps around histone proteins to form nucleosomes, creating chromatin structure. When chromatin is tightly packed (heterochromatin), genes are inaccessible to transcription factors and RNA polymerase. When loosely packed (euchromatin), genes are accessible for transcription. (2 marks) (2 marks)

Part b: Describe how histone modifications regulate transcription without altering DNA sequence.

Model Answer: Chemical modifications like acetylation or methylation of histone tails change how tightly DNA binds to histones. Acetylation typically loosens DNA-histone binding, making genes more accessible for transcription. This epigenetic regulation controls gene expression without changing the underlying DNA sequence. (2 marks) (2 marks)

Part c: State one advantage of epigenetic regulation over permanent DNA changes.

Model Answer: Epigenetic changes are reversible, allowing cells to adjust gene expression in response to environmental conditions without permanent genetic alterations. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for nucleosome structure, 1 mark for accessibility concept

Part b (2 marks): 1 mark for modification types, 1 mark for effect on transcription

Part c (1 mark): 1 mark for reversibility or flexibility advantage

Question 6

Part a: Explain how restriction enzymes create compatible DNA fragments for joining.

Model Answer: Restriction enzymes recognize and cut specific palindromic DNA sequences, often making staggered cuts that produce single-stranded overhangs called sticky ends. These sticky ends have complementary base sequences that can hydrogen bond with other fragments cut by the same enzyme. (2 marks) (2 marks)

Part b: Describe the role of DNA ligase in creating recombinant DNA molecules.

Model Answer: DNA ligase catalyzes phosphodiester bond formation between adjacent DNA fragments after their sticky ends have base-paired. This enzyme seals the sugar-phosphate backbone, creating a continuous recombinant DNA molecule combining DNA from different sources. (2 marks) (2 marks)

Part c: Identify one advantage of producing human insulin in bacteria versus extracting from animal sources.

Model Answer: Bacterial production yields human insulin identical to natural human insulin, eliminating immune reactions that can occur with animal-derived insulin. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for palindromic recognition, 1 mark for sticky ends

Part b (2 marks): 1 mark for phosphodiester bonds, 1 mark for joining fragments

Part c (1 mark): 1 mark for valid advantage

Question 7

Part a: Explain which type of chromosomal abnormality causes Turner syndrome.

Model Answer: Turner syndrome is caused by monosomy of the X chromosome (45,X karyotype). Affected individuals have only one X chromosome instead of the normal two sex chromosomes (XX or XY), resulting from nondisjunction during meiosis. (2 marks) (2 marks)

Part b: Explain why sex chromosome aneuploidies are more viable than autosomal aneuploidies.

Model Answer: Sex chromosome aneuploidies are more viable because X-inactivation compensates for dosage imbalances, and the Y chromosome contains fewer essential genes. Autosomal aneuploidies cause severe gene dosage imbalances affecting many essential functions, making most lethal except for small chromosomes. (2 marks) (2 marks)

Part c: State one physical characteristic commonly associated with Turner syndrome.

Model Answer: Short stature is a common characteristic, along with webbed neck, heart defects, or infertility. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for monosomy X, 1 mark for nondisjunction cause

Part b (2 marks): 1 mark for X-inactivation, 1 mark for gene dosage explanation

Part c (1 mark): 1 mark for valid characteristic

Question 8

Part a: Describe the three temperature-dependent steps of one PCR cycle.

Model Answer: Denaturation (94-96°C): High temperature breaks hydrogen bonds, separating DNA strands. Annealing (50-65°C): Temperature lowered so primers bind to complementary sequences on template strands. Extension (72°C): Taq polymerase synthesizes new DNA strands from primers in 5' to 3' direction. (2 marks) (2 marks)

Part b: Explain why PCR produces exponential amplification of the target sequence.

Model Answer: Each cycle doubles the amount of target DNA because both original strands and newly synthesized strands serve as templates in subsequent cycles. After n cycles, one molecule becomes 2^n molecules, creating exponential amplification (1→2→4→8→16...). (2 marks) (2 marks)

Part c: State why Taq polymerase is used instead of human DNA polymerase.

Model Answer: Taq polymerase is heat-stable and remains active at the high temperatures required for DNA denaturation, while human polymerase would denature. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for temperatures, 1 mark for describing each step

Part b (2 marks): 1 mark for doubling concept, 1 mark for exponential formula

Part c (1 mark): 1 mark for heat stability

Question 9

Part a: Explain what causes a frameshift mutation and how it affects the reading frame.

Model Answer: Frameshift mutations result from insertions or deletions of nucleotides that are not multiples of three. This shifts the reading frame for all codons downstream of the mutation, changing every amino acid from that point forward in the protein sequence. (2 marks) (2 marks)

Part b: Compare the protein-level effects of frameshift versus missense mutations.

Model Answer: Frameshift mutations alter all amino acids downstream, usually creating nonfunctional proteins or premature stop codons. Missense mutations change only one amino acid, which may have minimal effect if the substitution is similar or occurs in a non-critical region. (2 marks) (2 marks)

Part c: State why insertion or deletion of exactly three nucleotides is less damaging.

Model Answer: Three nucleotides equal one codon, so the reading frame remains intact, only adding or removing one amino acid while preserving the rest of the protein. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for indel cause, 1 mark for reading frame shift

Part b (2 marks): 1 mark for frameshift effects, 1 mark for missense comparison

Part c (1 mark): 1 mark for maintaining reading frame

Question 10

Part a: Identify the type of dominance exhibited and explain how it differs from complete dominance.

Model Answer: This shows incomplete dominance where neither allele is completely dominant. Heterozygotes express an intermediate phenotype (pink) rather than the dominant phenotype seen in complete dominance. Both alleles contribute to the phenotype creating a blended appearance. (2 marks) (2 marks)

Part b: Predict the phenotypic ratio if two pink snapdragons are crossed.

Model Answer: Pink $\times$ Pink ($Rr \times Rr$) produces: 1 RR (red) : 2 Rr (pink) : 1 rr (white), giving a 1:2:1 phenotypic ratio. This differs from the 3:1 ratio in complete dominance because heterozygotes have a distinct intermediate phenotype. (2 marks) (2 marks)

Part c: State the genotypic ratio from this cross.

Model Answer: The genotypic ratio is 1:2:1 (1 RR : 2 Rr : 1 rr), which equals the phenotypic ratio in incomplete dominance. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for identifying incomplete dominance, 1 mark for explaining intermediate phenotype

Part b (2 marks): 1 mark for correct ratio, 1 mark for showing cross

Part c (1 mark): 1 mark for 1:2:1 genotypic ratio

Question 11

Part a: Explain the process by which DNA information is converted to RNA.

Model Answer: Transcription converts DNA to RNA when RNA polymerase reads the DNA template strand (3' to 5') and synthesizes complementary RNA (5' to 3'). RNA polymerase substitutes uracil for thymine, creating an mRNA molecule carrying the genetic code. (2 marks) (2 marks)

Part b: Describe how RNA information becomes protein structure.

Model Answer: Translation occurs at ribosomes where mRNA codons are read in triplets. tRNA molecules with complementary anticodons deliver specific amino acids. Ribosomes catalyze peptide bond formation between amino acids, building a polypeptide chain according to the mRNA sequence. (2 marks) (2 marks)

Part c: Identify one exception to the central dogma's unidirectional flow.

Model Answer: Reverse transcription in retroviruses creates DNA from RNA templates, reversing the typical flow of genetic information. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for RNA polymerase function, 1 mark for complementary synthesis

Part b (2 marks): 1 mark for codon reading, 1 mark for amino acid assembly

Part c (1 mark): 1 mark for reverse transcription or valid exception

Question 12

Part a: Explain why triploid organisms have difficulty undergoing meiosis.

Model Answer: Triploid organisms have three sets of homologous chromosomes that cannot pair properly during meiosis I. With three homologs, proper segregation is impossible as chromosomes cannot divide evenly, leading to unbalanced gametes with varying chromosome numbers. (2 marks) (2 marks)

Part b: Describe how seedless watermelons are produced despite being sterile.

Model Answer: Diploid plants are crossed with tetraploid plants, producing triploid offspring. The diploid parent contributes $n=11$ gametes while the tetraploid contributes $2n=22$ gametes, resulting in $3n=33$ triploid plants that develop fruit but cannot produce viable seeds. (2 marks) (2 marks)

Part c: State one advantage of polyploidy in crop plants.

Model Answer: Polyploid plants often show increased vigor, larger fruit size, or enhanced stress tolerance compared to diploid varieties. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for pairing problem, 1 mark for segregation issues

Part b (2 marks): 1 mark for cross description, 1 mark for chromosome contributions

Part c (1 mark): 1 mark for valid advantage

Question 13

Part a: Explain why genes on the same chromosome tend to be inherited together.

Model Answer: Linked genes are physically connected on the same chromosome and move as a unit during meiosis. They cannot assort independently like genes on different chromosomes because they travel together unless separated by crossing over. (2 marks) (2 marks)

Part b: Describe how recombination frequency relates to the distance between linked genes.

Model Answer: Recombination frequency increases with distance between genes. Genes far apart have more opportunities for crossing over between them, while closely linked genes rarely separate. One map unit equals 1% recombination frequency, allowing genetic mapping of relative positions. (2 marks) (2 marks)

Part c: State the maximum recombination frequency possible between two linked genes.

Model Answer: 50% recombination frequency is maximum, occurring when genes are far apart or on different chromosomes, appearing to assort independently. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for physical linkage, 1 mark for inheritance together

Part b (2 marks): 1 mark for distance relationship, 1 mark for map units

Part c (1 mark): 1 mark for 50% maximum

Question 14

Part a: Explain the structure and function of telomeres at chromosome ends.

Model Answer: Telomeres are repetitive DNA sequences (TTAGGG in humans) at chromosome ends that protect coding DNA from degradation during replication. They act as disposable buffers, preventing loss of essential genetic information when DNA polymerase cannot replicate chromosome ends completely. (2 marks) (2 marks)

Part b: Describe why telomeres shorten with each replication cycle.

Model Answer: DNA polymerase cannot synthesize the very end of the lagging strand because it requires a primer ahead of it. When the final RNA primer is removed, a gap remains that cannot be filled, causing progressive telomere shortening with each division. (2 marks) (2 marks)

Part c: Identify which enzyme can extend telomeres and where it is typically active.

Model Answer: Telomerase extends telomeres by adding repetitive sequences; it's active in germ cells, stem cells, and cancer cells but not most somatic cells. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for repetitive structure, 1 mark for protective function

Part b (2 marks): 1 mark for end replication problem, 1 mark for primer gap

Part c (1 mark): 1 mark for telomerase and activity sites

Question 15

Part a: Explain how the lac operon is regulated when lactose is absent.

Model Answer: When lactose is absent, the repressor protein binds to the operator region, blocking RNA polymerase from transcribing the lac genes. This prevents unnecessary production of lactose-digesting enzymes when the substrate is unavailable, conserving cellular resources. (2 marks) (2 marks)

Part b: Describe what happens to lac operon expression when lactose is present.

Model Answer: Lactose (allolactose) acts as an inducer, binding to the repressor protein and causing a conformational change. The repressor releases from the operator, allowing RNA polymerase to transcribe genes for lactose metabolism enzymes (β -galactosidase, permease, transacetylase). (2 marks) (2 marks)

Part c: State the advantage of inducible gene expression for bacterial cells.

Model Answer: Inducible expression conserves energy and resources by producing enzymes only when their substrate is available, increasing efficiency. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for repressor binding, 1 mark for blocking transcription

Part b (2 marks): 1 mark for inducer binding, 1 mark for allowing transcription

Part c (1 mark): 1 mark for efficiency advantage

Question 16

Part a: Explain how guide RNA directs Cas9 to specific DNA targets.

Model Answer: Guide RNA contains a sequence complementary to the target DNA site. Through base pairing, it guides Cas9 enzyme to the specific genomic location where the guide RNA matches the DNA sequence, ensuring precise targeting for cutting. (2 marks) (2 marks)

Part b: Describe two ways cells can repair DNA after Cas9 creates a double-strand break.

Model Answer: Non-homologous end joining (NHEJ) quickly rejoins broken ends but often introduces insertions/deletions causing knockouts. Homology-directed repair (HDR) uses a template to precisely repair the break, allowing specific sequence insertion or correction. (2 marks) (2 marks)

Part c: State one medical application of CRISPR-Cas9 technology.

Model Answer: CRISPR can correct genetic mutations causing diseases like sickle cell anemia by editing the defective gene sequence in patient cells. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for complementary sequence, 1 mark for targeting mechanism

Part b (2 marks): 1 mark for NHEJ, 1 mark for HDR

Part c (1 mark): 1 mark for valid medical application

Question 17

Part a: Explain why X-linked recessive traits affect males more than females.

Model Answer: Males have only one X chromosome (hemizygous), so a single recessive allele causes expression of the trait. Females have two X chromosomes and need two recessive alleles for expression, making them less likely to be affected since they can be carriers with one normal allele. (2 marks) (2 marks)

Part b: Predict the offspring ratios when a carrier female crosses with a normal male.

Model Answer: Carrier female ($X^H X^h$) $\times$ Normal male ($X^H Y$) produces: 25% normal female ($X^H X^H$), 25% carrier female ($X^H X^h$), 25% normal male ($X^H Y$), 25% affected male ($X^h Y$). Thus 50% of males are affected while no females express the disorder. (2 marks) (2 marks)

Part c: State why an affected male cannot pass the disorder to his sons.

Model Answer: Males pass their Y chromosome to sons and X chromosome to daughters, so X-linked traits cannot pass from father to son. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for hemizygous explanation, 1 mark for female carrier concept

Part b (2 marks): 1 mark for correct ratios, 1 mark for showing cross

Part c (1 mark): 1 mark for Y chromosome inheritance

Question 18

Part a: Explain why mitochondrial inheritance follows a maternal pattern.

Model Answer: The egg cell contributes nearly all cytoplasm to the zygote, including mitochondria and their DNA. Sperm contribute primarily nuclear DNA with minimal cytoplasm, and any paternal mitochondria are typically degraded after fertilization, resulting in maternal-only inheritance. (2 marks) (2 marks)

Part b: Describe how mitochondrial DNA structure differs from nuclear DNA.

Model Answer: Mitochondrial DNA is circular like bacterial DNA, much smaller than nuclear chromosomes, and lacks histones for packaging. It encodes essential components for electron transport and ATP synthesis, reflecting the endosymbiotic origin of mitochondria. (2 marks) (2 marks)

Part c: State one application of mitochondrial DNA analysis in human studies.

Model Answer: Mitochondrial DNA is used to trace maternal lineages in evolutionary studies or identify remains in forensic investigations. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for egg cytoplasm contribution, 1 mark for sperm exclusion

Part b (2 marks): 1 mark for circular structure, 1 mark for size/package differences

Part c (1 mark): 1 mark for valid application

Question 19

Part a: Explain why human genes must be modified before expression in bacteria.

Model Answer: Human genes contain introns that bacteria cannot remove because they lack splicing machinery. The cDNA version (complementary DNA made from mRNA) lacks introns and can be directly translated in bacteria. Additionally, bacterial promoters must be added for transcription. (2 marks) (2 marks)

Part b: Describe how plasmid vectors facilitate gene transfer into bacterial cells.

Model Answer: Plasmids are circular DNA molecules that replicate independently in bacteria. The human gene is inserted into the plasmid using restriction enzymes and ligase, then the recombinant plasmid is introduced into bacteria through transformation where it replicates with each cell division. (2 marks) (2 marks)

Part c: Identify one advantage of using bacteria for protein production.

Model Answer: Bacteria reproduce rapidly (every 20 minutes), allowing quick scale-up of production to generate large quantities of protein economically. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for intron issue, 1 mark for cDNA solution

Part b (2 marks): 1 mark for plasmid properties, 1 mark for transformation process

Part c (1 mark): 1 mark for rapid reproduction advantage

Question 20

Part a: Explain how DNA compaction allows 2 meters of DNA to fit in a 10-micrometer nucleus.

Model Answer: DNA wraps around histone octamers forming nucleosomes, which coil into 30nm chromatin fibers. These fibers loop and condense further through scaffold proteins, ultimately forming the highly condensed metaphase chromosomes visible during cell division, achieving 10,000-fold compaction. (2 marks) (2 marks)

Part b: Describe the relationship between chromatin structure and gene expression.

Model Answer: Heterochromatin (tightly packed) silences genes by preventing transcription factor access, while euchromatin (loosely packed) allows active transcription. Chromatin remodeling complexes and histone modifications dynamically regulate which regions are accessible, controlling gene expression patterns. (2 marks) (2 marks)

Part c: State when chromosomes are most condensed during the cell cycle.

Model Answer: Chromosomes are most condensed during metaphase of mitosis when they align at the cell equator before separation. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for hierarchical packaging, 1 mark for compaction level

Part b (2 marks): 1 mark for hetero/euchromatin distinction, 1 mark for accessibility concept

Part c (1 mark): 1 mark for metaphase